LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 2 Test (Set B): Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

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SECTION A

Q1. Which of the following gives carbon dioxide with dilute hydrochloric acid? **[1]**

Answer: (c) Sodium hydrogencarbonate. $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$

Q2. Which of these has the highest pH? **[1]**

Answer: (b) Milk of magnesia (it is basic)

Q3. Which indicator turns pink in a basic solution? **[1]**

Answer: (d) Phenolphthalein

Q4. The chemical formula of Plaster of Paris is: **[1]**

Answer: (b) $CaSO_4 \cdot \frac{1}{2}H_2O$

Q5. The stinging hairs of nettle leaves inject: **[1]**

Answer: (d) methanoic acid

Q6. **Assertion (A):** Tooth decay starts when the pH in the mouth falls below 5.5. **Reason (R):** Below pH 5.5, tooth enamel starts to get corroded. **[1]**

Answer: (a)

SECTION B

Q7. What is a universal indicator? Why is it more useful than litmus? **[2]**

Answer: A universal indicator is a mixture of several indicators that shows different colours at different pH values (different concentrations of H^+ ions). Litmus only tells whether a solution is acidic or basic. A universal indicator also tells how strong the acid or base is (its pH).

- Definition: 1
- Shows strength / pH: 1

OR, answer: HCl ionises completely in water and produces many more H^+ ions. Acetic acid ionises only partially, so it produces fewer H^+ ions.

- HCl ionises completely: 1
- Acetic acid partially: 1

Q8. Why should Plaster of Paris be stored in a moisture-proof container? **[2]**

Answer: Plaster of Paris reacts with water to form gypsum ($CaSO_4 \cdot 2H_2O$), which sets into a hard solid mass. If moisture gets in, the powder becomes hard and useless.

- Reacts with water to form gypsum: 1
- Sets hard / becomes useless: 1

SECTION C

Q9. A solution has a pH of 3. **[3]**

(i) Acidic [1]

(ii) The pH increases towards 7, because the H^+ ion concentration decreases. [1]

(iii) The pH rises until it reaches 7 (neutral). [1]

Q10. How is baking powder different from baking soda? Why does baking powder make a cake soft and spongy? **[3]**

Answer: Baking soda is only sodium hydrogencarbonate. Baking powder is a mixture of baking soda and a mild edible acid such as tartaric acid. When it is heated or mixed with water, it releases carbon

dioxide gas, which makes the dough rise, so the cake becomes soft and spongy. The acid also stops the bitter taste of sodium carbonate.

- Difference: 1
- CO_2 released: 1
- Makes it rise / soft: 1

OR, answer: (i) $HCl + NaOH \rightarrow NaCl + H_2O$ (sodium chloride) (ii) $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$ (potassium sulphate) (iii) $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$ (calcium nitrate)

- Each equation + salt: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Lime water turns milky because insoluble calcium carbonate forms: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$. With excess CO_2, the milkiness disappears because soluble calcium hydrogencarbonate forms: $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ [2]

- Each observation + equation: 1

(ii) (Any three) Antacids such as milk of magnesia neutralise excess acid in the stomach during indigestion. Basic toothpaste neutralises the acid produced by bacteria in the mouth, preventing tooth decay. Baking soda (a mild base) is applied on a bee sting to neutralise the acid. Quicklime or slaked lime is added to acidic soil. [3]

- Any three examples with explanation: 1 each

OR, answer: (a) Water of crystallisation is the fixed number of water molecules present in one formula unit of a salt crystal. Copper sulphate crystals ($CuSO_4 \cdot 5H_2O$) are blue. On heating they lose their water of crystallisation and turn white (anhydrous $CuSO_4$), and water droplets appear in the test tube. When water is added, the blue colour returns. (b) (i) Washing soda (sodium carbonate). (ii) Bleaching powder.

- (a) definition 1, heating 1, adding water 1
- (b) 1 + 1

SECTION E

Q12. Plants grow well only when the soil pH is in a suitable range. A farmer found that the soil in his field had a pH of 5. In his area, rain water… **[4]**

(i) Acidic (pH less than 7). [1]

(ii) Quicklime (calcium oxide), slaked lime (calcium hydroxide) or chalk (calcium carbonate): a base that neutralises the acid. [1]

(iii) Aquatic animals can survive only in a narrow pH range, so a lower pH makes survival difficult for river life. Plants need a specific pH range for healthy growth, so the wrong soil pH reduces crop yield. [2]

- River life: 1
- Soil pH for plants: 1